

INTERNSHIP-CONNECT

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Project Report

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Limitations

- One student can generally have only one offline internship per company, although multiple online internships per company may be allowed.
- The platform's effectiveness depends on accurate and authentic data from participating companies and students, which can sometimes pose challenges.
- Relationship-building and social interaction among interns, which are critical parts of internships, may be limited in virtual or digitally mediated environments and require deliberate intervention
- Even though this project is meant to make it easy for students to find proper and right internship it very likely that not all students will be assisted or find places due to the high number of individuals

1.5 DEVELOPMENT METHODOLOGY

Agile Methodology

Agile methodology is adopted for the development of the Intern Connect platform due to its iterative and flexible nature. By dividing the project into short development cycles or sprints, Agile allows for the continuous delivery of functional components, enabling frequent feedback from users and stakeholders. This approach ensures timely adaptations to changing requirements, improving overall product quality and user satisfaction. Agile fosters collaboration among the development team and stakeholders, promoting transparency and efficient communication. For Intern Connect, Agile supports prioritizing key features, rigorous testing, and incremental enhancements, making it ideal for building a scalable, user-focused internship management system.

For developing an internship portal like Intern Connect, the recommended methodology is typically a structured software development process that ensures clarity, completeness, and iterative progress. One commonly applied approach is the Waterfall method, which proceeds through sequential phases such as requirements analysis, system design, implementation, testing, and maintenance. This method emphasizes a clear understanding of system needs upfront,

followed by careful design using ER diagrams and use case diagrams to model data relationships and user interactions. After implementation, thorough testing ensures the system meets specifications before deployment.

Agile methodology is an iterative and incremental approach to software development that emphasizes flexibility, collaboration, and customer feedback. It breaks the development process into small, manageable units called sprints or iterations, typically lasting two to four weeks. Each sprint results in a working product increment that can be reviewed and tested by stakeholders. This frequent delivery allows teams to quickly adapt to changing requirements, improve features based on user feedback, and address issues promptly.

Agile promotes close communication among cross-functional teams, including developers, designers, testers, and business stakeholders, fostering a collaborative environment. The methodology values working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a rigid plan.

Agile practices often include daily stand-up meetings, sprint planning, retrospectives, and continuous integration to maintain transparency and ensure steady progress

CHAPTER 2

LITERATURE REVIEW

The transition from higher education to the professional workforce is a universal challenge, yet it is particularly acute in Zambia, where high rates of youth unemployment persist despite educational attainment. This discrepancy is often attributed to an "educational mismatch," where the theoretical knowledge acquired in academic institutions fails to align with the practical skills demanded by the local labor market (Chimfwembe & Nyirongo, 2019). Within this context, internships are widely recognized as a critical pedagogical bridge, offering students the experiential learning necessary to translate theory into practice and enhance their employability (Mokhtar, 2023).

However, the current ecosystem for connecting students with these vital opportunities in Zambia is fragmented and inefficient. Existing solutions, ranging from general job boards to international professional networks, are ill-suited to the specific needs of students and recent graduates, often leaving them overlooked in a crowded and competitive landscape. Similarly, employers struggle to efficiently identify and recruit qualified entry-level talent from this demographic.

This literature review synthesizes empirical and theoretical research to establish a comprehensive understanding of the role of internships and the technological landscape designed to facilitate them. The review is divided into two primary sections. The first, **Related Work**, examines scholarly research on the impact of internships on skills development and employability, with a specific focus on the Zambian context. The second, **Previous Existing Systems**, evaluates the technological platforms and frameworks that have been developed to manage internship processes. By analyzing both the scholarly discourse and the technological solutions, this review identifies critical gaps in access, implementation, and support that the proposed Internship Connect platform aims to address.

Review

Related Work

The value of internships as a tool for career readiness is well-established in academic literature. Meyers and Moyo (2021) provide a critical, Zambia-specific analysis in their study, "Effectiveness of Internships as Pedagogical Practices in Promoting Employability Skills amongst Graduating Students in Selected Social Science Degree Programmes in Zambia." Their research moves beyond general claims by investigating how internships within social science degrees—often considered less vocational—foster concrete, transferable skills like leadership, problem-solving, and ethical professionalism. The study emphasizes that real-world organizational exposure is the key mechanism for transforming theoretical knowledge into practical, employable attributes, thereby validating the need for structured experiential learning within the Zambian higher education system. A significant gap identified by their work is not the *value* of internships, but rather the systemic challenge of *access and quality*. Their findings imply that not all students have equal opportunity to participate in meaningful internships, highlighting the need for a centralized, transparent marketplace that can democratize access, particularly for students in fields traditionally overlooked by corporate recruiters.

Building on this conceptual foundation, Mokhtar (2023) provides a strong theoretical framework in "Internship Program: A Bridge to Close the Gap Between Theory and Practice." The article explicitly defines internships as a fundamental pedagogical bridge designed to integrate academic learning with real-world application. Mokhtar argues that structured internships should be considered essential curricular components, not merely optional work placements, as they allow students to contextualize theoretical knowledge. This work is crucial as it directly names and addresses the pervasive theory-practice gap that hinders graduate employability. However, while Mokhtar effectively establishes the *why* of internships, the article does not delve into the *how* of efficiently creating, managing, and scaling these vital bridges, particularly within resource-constrained environments like Zambia. The gap, therefore, lies in the implementation mechanism—a gap that a dedicated technological platform can fill by providing the necessary infrastructure to systematically build and support these connections.

Further focusing on outcomes, Bawica (2021) examines "The University Internship Program and its Effects on Students' Employability Readiness." This research measures how university-led internship programs impact a student's preparedness for the job market, assessing dimensions such as practical skill competency, professional networking ability, and confidence in job-seeking activities. While Bawica's research effectively measures the positive effects of

internships on readiness, it may not fully address the operational and logistical challenges universities face in setting up and managing these programs at scale. The gap lies in the infrastructure required to consistently deliver high-quality internship experiences to all students. A platform that streamlines the entire process—from connecting with a wide network of employers to managing applications and tracking student progress—could enhance an institution's capacity to achieve the employability readiness Bawica describes.

Empirically grounding the problem in the Zambian context, Chimfwembe and Nyirongo (2019) present a compelling case in "Internships and Educational Mismatch among Youths in Zambia: Evidence from the School-to-Work Transition Survey." Using national survey data, the authors demonstrate that internships serve as a powerful corrective measure to the skills mismatch, helping to align academic knowledge with practical market demands and thereby reducing youth unemployment and underemployment. Their research identifies the core problem but does not propose a scalable solution for connecting the entire ecosystem. The identified gap is the lack of a dedicated channel that efficiently connects the supply of students from diverse academic backgrounds with the demand for specific skills from employers, particularly small and medium enterprises (SMEs). A specialized platform like Internship Connect is designed to be this very channel, directly targeting the educational mismatch by facilitating connections that are currently informal or inefficient.

Previous Existing Systems

Beyond scholarly research, evaluating existing technological solutions reveals significant opportunities for a tailored platform. Myndyti and Kadriu (2020) describe the development of a dedicated Internship Management System (IMS) in North Macedonia in their work, "Developing an internship management system for enhancing collaboration between academia and industry." Their system focuses on improving administrative coordination between universities and employers through modules for task management, reporting, and notifications. Built on specific web technologies like MVC frameworks and MySQL, it represents a trend towards customized, database-driven platforms. However, a key limitation is that such systems are typically closed ecosystems designed for managing *existing* institutional partnerships *after* a placement has been made. They do not function as open, discoverable marketplaces for students to find new

opportunities. This creates a gap for a public-facing discovery and connection tool that could later integrate with formal institutional IMS.

Similarly, the Knowledge Serve Intern Management System (2023), as documented in the IRJMETS journal, represents a more advanced, feature-rich IMS. It includes real-time progress monitoring, attendance tracking, and automated report generation, built with C# and the .NET framework. This system is aimed at improving transparency and performance analysis for administrators and trainers, focusing heavily on oversight and evaluation. Like the system by Myndyti and Kadriu (2020), this is primarily an administrative-backend solution. The gap remains the lack of a student-centric frontend designed for exploration, networking, and personal brand building. A platform that prioritizes the user experience for the student and employer first, rather than starting from an administrative perspective, would address this unmet need.

Shifting to public platforms, LinkedIn is the world's largest professional network. While highly effective for experienced professionals, it presents significant challenges for students and fresh graduates. The platform is intensely competitive, often requiring an established network to get noticed. Job postings are frequently geared towards candidates with several years of experience, and its algorithms tend to favor established profiles, making it difficult for entry-level talent to stand out. The gap left by LinkedIn is a platform dedicated to the "beginner" segment of the job market—a less competitive space focused exclusively on internships, part-time jobs, and entry-level roles where features can be tailored to showcase academic projects and potential, rather than just prior work experience.

Domestically, platforms like Jobs Zambia, GoZambiaJobs, and Jobsbwana are the leading online job boards in Zambia. They primarily function as digital bulletin boards where employers post vacancies. These generalist platforms are dominated by advertisements for roles requiring 3-5 years of experience, effectively sidelining students and graduates. They lack functionality for structuring internship programs, managing applications for short-term roles, or fostering mentor connections. They are essentially bulletin boards, not integrated career development ecosystems. The gap, therefore, is a platform dedicated to curating opportunities for the "beginner" segment, filtering out the noise of senior-level positions and providing a tailored experience.

A more direct comparator is the Kubona Internship Network (2024), which is specifically focused on internships in the region. Its stated aim of "Bridging Skills and Cultures Through

Internships" suggests a focus on both local and international opportunities, emphasizing cultural exchange. While Kubona addresses the internship niche, the Internship Connect platform can differentiate itself through a hyper-local, partnership-driven approach. By building features that directly address the findings of Zambian-specific research (Meyers & Moyo, 2021; Chimfwembe & Nyirongo, 2019), such as tracking skill acquisition against local employer needs and providing analytics to universities on student placement outcomes, Internship Connect can offer a deeper, more integrated solution for the Zambian market.

Conclusions

The literature consistently affirms the critical role of internships in bridging the gap between academic preparation and professional employment, a challenge particularly relevant in Zambia. However, a significant disconnect exists between the recognized value of these experiences and the mechanisms available to facilitate them efficiently and at scale. Scholarly research highlights systemic issues of access and operational logistics, while an analysis of existing technological systems reveals a landscape of fragmented solutions that are either too administrative, too general, or too competitive to effectively serve Zambian students and employers.

The collective findings from both bodies of work demonstrate a clear and compelling need for an integrated, student-centric platform. Such a platform must function as a centralized hub that not only connects students with relevant opportunities through intelligent matching but also supports the entire ecosystem students, universities, and employers in a manner that is tailored to the local context. The Internship Connect project is conceived to address these identified gaps directly, proposing a technological solution that is both pedagogically informed and designed for the specific realities of the Zambian entry-level job market.